Project 1 - Sensor Characterization

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***Abstract*—This paper details the design, implementation, and characterization of an instrumentation system for a resistive force sensor. A Wheatstone quarter-bridge configuration was utilized to convert mechanical force into differential voltage, which was subsequently processed through an instrumentation amplifier for signal conditioning. The system was characterized across a 0–500 g range, achieving a peak sensitivity of 6.51 mV/g. Experimental analysis was performed to evaluate common-mode rejection ratio (CMRR), system resolution, and thermal effects, revealing an average temperature coefficient of 1.37%/°C. Furthermore, the efficacy of a half-bridge configuration in mitigating thermal noise was investigated. The results demonstrate that while the system provides a reliable linear response within the 30–200 g range, factors such as mechanical creep, resistor tolerances, and thermal sensitivity necessitate specific design trade-offs in high-precision applications.**

I. Introduction

Force sensors are widely used in instrumentation systems to measure mechanical load, pressure, and weight in a variety of industrial and research applications. Many force sensing devices operate by converting an applied mechanical stimulus into a change in electrical resistance. Because the resulting resistance change is often very small, bridge circuits and signal conditioning electronics are required to accurately measure the sensor output.

One of the most common configurations used for resistive sensors is the Wheatstone bridge. The Wheatstone bridge converts small resistance variations into a differential voltage that can be measured with high precision. In a quarter-bridge configuration, a single sensing element is combined with three reference resistors to form the bridge network. When the bridge is balanced, the differential output voltage is ideally zero, and any change in the sensor resistance due to an applied force produces a measurable voltage difference. The sensitivity of the bridge depends on the resistor values and the excitation voltage applied to the circuit.

Because the output signal of a bridge circuit is typically very small, amplification and filtering stages are required before the signal can be used for measurement or digitization. Differential amplifiers or instrumentation amplifiers are commonly used to increase the signal amplitude while rejecting common-mode noise. Additional signal conditioning, such as low-pass filtering, is often required to reduce high-frequency noise and improve measurement stability.

Despite careful circuit design, several factors can introduce error into sensor measurements. Temperature variations can alter the resistance of both the sensing element and the bridge resistors, causing drift in the output voltage. Imperfect resistor tolerances can also limit the balance of the bridge and reduce common-mode rejection. Additionally, electrical noise and finite amplifier resolution can limit the smallest detectable change in force.

In this experiment, a Wheatstone quarter bridge was designed to interface with a resistive force sensor. The bridge output was amplified using a signal conditioning circuit in order to characterize the sensor response over a range of applied forces. The system was then used to investigate important instrumentation properties including sensitivity, calibration accuracy, common-mode rejection ratio (CMRR), temperature effects, and measurement resolution.

II. Method

A. Sensor design

1) Wheatstone quarter bridge design: A small test bench was designed to hold the sensor, heating pad, and breadboards together so that the sensor remained in the same position for each test. To determine the value of R_0, a 100 g weight was placed directly on the sensor and the resulting resistance was measured. Initially, the value of R_0 fluctuated between approximately 30 kΩ and 9 kΩ. This behavior was believed to be caused by the edges of the weight pinching the sensor. To mitigate this issue, a piece of corrugated cardboard was attached to the sensor using two loops of electrical tape, and a reference mark was added to ensure the weight was placed in the same location for each measurement. The cardboard distributed the load more evenly across the sensor, resulting in a consistent reading of approximately 3.3 kΩ.

A Wheatstone quarter-bridge circuit was constructed with the sensor in the lower-right leg. Resistors of 3.3 kΩ were used for the remaining bridge elements so that the bridge was balanced with a 100 g load on the sensor. To determine the sensitivity of the circuit, weights ranging from 30 g to 500 g were applied, and the output voltage was recorded at 30 s, 60 s, and 120 s to account for potential settling time. As shown in Table IV, the sensitivity increased from 6.51 mV/g to 5.34 mV/g when an additional 90 s of settling time was allowed.

2) Amplifier design: An instrumentation amplifier was used to offset the bridge output by 2.5 V. The bridge produced approximately -2.5 V at 0 g and 1.15 V at 500 g, yielding a span of 3.65 V, or 73% of full scale. Because the output already exceeded 70% of full scale, no additional gain was applied. Instead, the amplifier introduced a 2.5 V offset so that

the theoretical output range became 0–3.65 V. A potentiometer configured as a voltage divider was used to adjust the DC offset, allowing the system to be tuned back to 2.5 V at 100 g as needed.

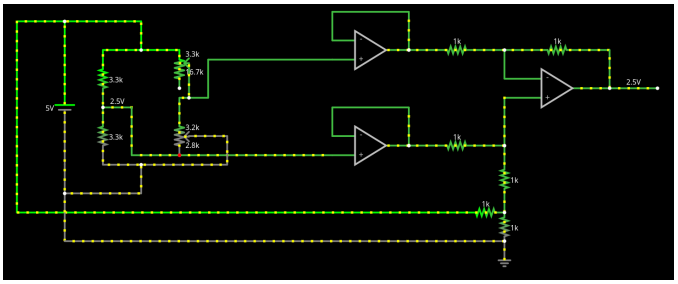

Fig. 1. Schematic – Wheatstone bridge and amplifier

3) Sensor calibration curve: To evaluate the complete sensor–bridge–amplifier system, ten weights from 0 g to 500 g were applied. The measured output range was 1.1–3.9 V, and the resulting calibration curve is shown in Fig. 2. The response was approximately linear between 30 g and 200 g. Data beyond 200 g were inconsistent due to the physical difficulty of stacking the weights reliably.

B. Thermal effect

1) Temperature coefficient: : To measure the temperature coefficient of the sensor, the heating pad was initially powered at 5 V and later increased to 9 V to accelerate heating. As the heating pad warmed, the output voltage for 50 g, 100 g, and 200 g loads was recorded at two different temperatures. To determine the temperature rejection ratio, the same procedure was repeated using a full-bridge configuration, with an additional sensor placed in the upper-right leg and loaded with 100 g to balance the bridge. These measurements were used to calculate the thermal noise rejection ratio and the temperature coefficient.

Temperature (°C)	Resistance ($k\Omega$)	Force (g)
39	3.6	100
45	3.5	100
50	3.0	100
63	2.4	100

TABLE I
MEASURED RESISTANCE VERSUS TEMPERATURE AT CONSTANT FORCE.

Parameter	Value
Slope (dR/dT)	$-0.0528\ k\Omega/°C$
Reference Resistance	$3.6\ k\Omega$
Temperature Coefficient	$-0.0147\ /°C$
Percent Change	$-1.47\%/°C$

TABLE II
CALCULATED TEMPERATURE COEFFICIENT OF RESISTANCE.

2) Wheatstone half bridge thermal error:

C. Sensor system

1) Common Mode Rejection Ratio (CMRR): To measure the common-mode rejection ratio (CMRR) of the complete system, the bridge was powered with a 10 Hz, 260 V_{PP}, 5 V offset sinusoidal signal. Care was taken to ensure that the amplifiers were not powered by this signal. With a 100 g load on the sensor, the output consisted of a 2.6 V offset and a 10 Hz, 100 V_{PP} sinusoidal component.

2) Resolution: To determine the system resolution, a 50 g weight was placed on the sensor and the output voltage was recorded. A 1 g weight was then added on top of the 50 g load and the voltage was recorded again. The 1 g weight was removed and the voltage was recorded once more. This sequence was repeated ten times. The procedure was then repeated using a 5 g weight in place of the 1 g weight.

III. RESULTS

A. Wheatstone quarter bridge resistor values

The ideal resistor value is 3.3 kΩ

The measured resistor values are as follows:

Resistor	Actual kΩ	Position	Color
R01	3.26	Top-Left	Green
R02	3.27	Bottom-Left	Red
R03	3.25	Top-Right	None

TABLE III
RESISTOR VALUES

B. Bridge sensitivity

Weight(g)	Volts(30s)	Volts(60s)	Volts(120s)
100	25m	25m	25m
110	120m	127m	130m
120	200m	198m	215m
130	250m	270m	280m
150	365m	372m	380m
200	550m	567m	560m
300	877m	878m	885m
500	1.145	1.147	1.153

TABLE IV
SENSITIVITY TRIAL 1

Weight(g)	Volts(30s)	Volts(60s)	Volts(120s)
0	-2.494		
30	-2.166	-2.146	-2.133
50	-1.270	-1.234	-1.174
70	-570m	-561m	-533m
100	-60m	-53m	-46m
110	50m	61m	76m
120	145m	154m	164m
130	237m	249m	245m
150	370m	380m	383m
200	590m	600m	610m
300	875m	881m	888m
500	1.125	1.137	1.150

TABLE V
SENSITIVITY TRIAL 2

Time Interval	**Sensitivity (V/g)**	**Sensitivity (mV/g**	**Linearity (R^2)**
30 Seconds	0.00651	6.51	0.599
60 Seconds	0.00538	5.38	0.581
120 Seconds	0.00534	5.34	0.584

TABLE VI
CALCULATED SENSITIVITY AND LINEARITY ACROSS TIME INTERVALS

C. Low-pass filter

1) What is a reasonable cut-off frequency for the circuit?: 60 Hz is a reasonable cut-off frequency because the sensor system is designed to measure a constant force on the input, that is unlikely to change at a rate of 60+ Hz.

2) What influences the upper and lower limits of "reasonable"?: The size of the capacitor. The limited amount of input variance.

D. Instrumentation amplifier

1) 2 trade-offs vs. the differential amplifier:

A) By using an Instrumentation amplifier system, we are not concerned with the input impedance to the amplifier as the unity gain buffers act as a high impedance input to the differential amplifier.
B) A downside to the instrumentation amplifier is with 2 additional amplifiers the cost of components increases.

2) Stage gain distribution: The amplifier system was designed utilizing unity gain among all 3 amplifiers, with a 2.5 volt offset to the differential amplifier to ensure the output is always positive.

E. Calibration curve

Weight(g)	Volts
0	1.116
30	1.114
50	1.23
70	1.87
100	2.48
130	2.86
150	3.12
200	3.44
300	3.78
500	3.9

TABLE VII
AMPLIFIER OUTPUT MEASUREMENTS

Utilizing the calibration curve any weight from 0 - 500 grams can be estimated. The estimation will be more accurate for the 0-200g range where the curve can be approximated by a linear equation.

$$V_{\text{linear}} = 1.38 \times 10^{-2} W + 8.93 \times 10^{-1} \tag{1}$$

$$V_{\text{quadratic}} = -2.26 \times 10^{-5} W^2 + 1.74 \times 10^{-2} W + 8.23 \times 10^{-1} \tag{2}$$

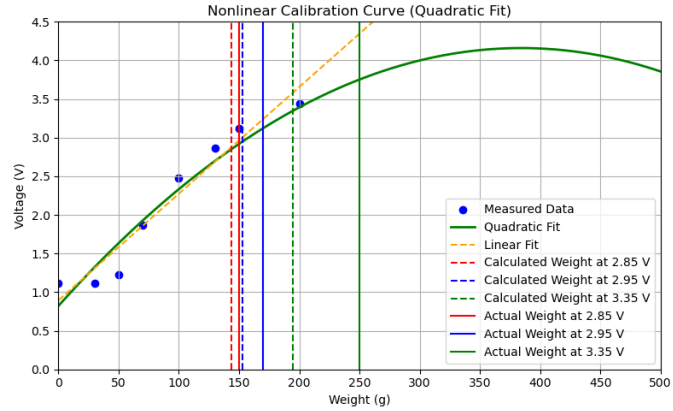

Fig. 2. Calibration Curve

F. Temperature co-efficient

Weight(g)	Temperature (°F)	Voltage (V)
50	33.2	1.61
50	45.3	2.02
100	39.2	2.77
100	46.1	2.96
200	37.9	3.45
200	47.8	3.72

TABLE VIII
TEMPERATURE MEASUREMENTS

The temperature coefficient (TC) for a sensor is calculated using the formula:

$$TC = \frac{\Delta V}{V_{\text{ref}}\, \Delta T} \tag{3}$$

where:

- V_{ref} is the voltage at the lower temperature,
- $\Delta V = V_{\text{high}} - V_{\text{low}}$,
- $\Delta T = T_{\text{high}} - T_{\text{low}}$.

Calculation for each weight: **50 g:**

$$\Delta V = 2.02 - 1.61 = 0.41\,\text{V}, \quad \Delta T = 45.3 - 33.2 = 12.1\,^\circ\text{C}$$

$$TC_{50} = \frac{0.41}{1.61 \times 12.1} \approx 0.0211 \text{ per °C } (2.11\%/\text{C})$$

100 g:

$$\Delta V = 2.96 - 2.77 = 0.19\,\text{V}, \quad \Delta T = 46.1 - 39.2 = 6.9\,^\circ\text{C}$$

$$TC_{100} = \frac{0.19}{2.77 \times 6.9} \approx 0.00996 \text{ per °C } (1.0\%/\text{C})$$

200 g:

$$\Delta V = 3.72 - 3.45 = 0.27\,\text{V}, \quad \Delta T = 47.8 - 37.9 = 9.9\,^\circ\text{C}$$

$$TC_{200} = \frac{0.27}{3.45 \times 9.9} \approx 0.0079 \text{ per °C } (0.79\%/\text{C})$$

Average Temperature Coefficient:

$$TC_{\text{avg}} = \frac{TC_{50} + TC_{100} + TC_{200}}{3}$$

$$TC_{\text{avg}} = \frac{0.0211 + 0.00996 + 0.0079}{3} \approx 0.0137 \,\text{per °C}\ (1.37\%/\text{C})$$

Weight (g)	Temperature (°C)	Voltage (V)	TC (°C)
50	33.2 – 45.3	1.61 – 2.02	2.11
100	39.2 – 46.1	2.77 – 2.96	1.00
200	37.9 – 47.8	3.45 – 3.72	0.79
Average			1.37

TABLE IX
TEMPERATURE COEFFICIENT CALCULATION FOR DIFFERENT WEIGHTS.

G. CMRR

- **Bridge CMRR:**

$$\text{CMRR}_{\text{bridge, dB}} = 20\log_{10}\left(\frac{1}{t}\right) = 20\log_{10}\left(\frac{1}{0.05}\right) \approx 26.02 \text{ dB}$$

- **Linear Conversion:**

$$\text{CMRR}_{\text{bridge}} = 10^{26.02/20} \approx 20$$

$$\text{CMRR}_{\text{amp}} = 10^{80/20} = 10,000$$

- **Total CMRR Calculation:**

$$\frac{1}{\text{CMRR}_{\text{total}}} = \frac{1}{20} + \frac{1}{10,000} = 0.0501$$

$$\text{CMRR}_{\text{total, dB}} = 20\log_{10}\left(\frac{1}{0.0501}\right) \approx 26.00 \text{ dB}$$

1) How will the amplifier affect the CMRR of the system if it's not ideal and has it's own CMRR of 80dB?: Based on the above calculations the amplifier with CMRR of 80dB results in an identical CMRR.

H. Resolution

1) What is the theoretical resolution of the amplifiers output?: The theoretical resolution of the amplifer output is infinitesimal as a fully analog system.

2) What factors may prevent the system from reaching its theoretical maximum?: Component noise is non ideal and absorbs some of the resolution. Sensor inaccuracy.

3) If the amplifier is connected to a 12-bit ADC that your system uses 70% of full-scale, what is the resolution in terms of force?:

$$V_{\text{ref}} = 3.5V$$

$$Resolution = \frac{V_{\text{ref}}}{2^{12}} = 0.85mV/bit$$

at ≈5.34 mV/g (see Table III-B)

$$Resolution = \frac{0.85}{5.34} = 0.159g/bit$$

I. Calibration weights

1) What is the mean and standard deviation of the voltage output in each case?: The mean and standard deviation can be found in the tables below

w/o weight (V)	with 5 g (V)	Δ (V)
1.23	1.50	0.27
1.25	1.44	0.19
1.36	1.56	0.20
1.28	1.40	0.12
1.22	1.43	0.21
1.21	1.44	0.23
1.19	1.43	0.24
1.21	1.48	0.27
1.21	1.47	0.26
1.22	1.52	0.30

TABLE X
RAW MEASUREMENTS WITH AND WITHOUT A 5 G LOAD.

Metric	Value	Units
Mean (w/o weight)	1.238	V
Std. Dev. (w/o weight)	0.049	V
Mean (with 5 g)	1.467	V
Std. Dev. (with 5 g)	0.049	V
Mean Sensitivity (Δ per 5 g)	0.229	V / 5 g
Std. Dev. (Δ)	0.053	V

TABLE XI
STATISTICAL SUMMARY AND MEASURED SENSITIVITY FOR A 5 G LOAD.

w/o weight (V)	with 1 g (V)	Δ (V)
1.20	1.31	0.11
1.21	1.32	0.11
1.22	1.31	0.09
1.22	1.35	0.13
1.21	1.30	0.09
1.21	1.26	0.05
1.21	1.31	0.10
1.23	1.32	0.09
1.22	1.28	0.06
1.21	1.28	0.07

TABLE XII
RAW MEASUREMENTS WITH AND WITHOUT A 1 G LOAD.

Metric	Value	Units
Mean (w/o weight)	1.214	V
Std. Dev. (w/o weight)	0.008	V
Mean (with 1 g)	1.304	V
Std. Dev. (with 1 g)	0.026	V
Mean Sensitivity (Δ per 1 g)	0.090	V / g
Std. Dev. (Δ)	0.024	V

TABLE XIII
STATISTICAL SUMMARY AND MEASURED SENSITIVITY FOR A 1 G LOAD.

2) Can the system accurately measure a difference of 1g/5g?: Yes, to some extent, the change in 1g/5g provided a noticeable difference in our sensor system. Though the sensitivity may make it difficult to identify the exact per gram difference.

J. Thermal noise rejection-ratio

Weight (g)	Temperature (°C)	Voltage (V)
50	32	1.20
50	65	1.57
100	32	2.25
100	65	2.53
200	32	3.06
200	65	3.20

TABLE XIV
HALF BRIDGE MEASUREMENTS

IV. CONCLUSION

A. Adjustments for a strain guage

Although the strain gauge sensors used provide a method for converting force into a resistance change, the developed system showed noticeable drift under constant loads. When a sustained force was applied, the resistance continued to drop and, as a result, the voltage differential to rise. Even after long settling times, the measured resistance continued to drift. This observation suggests the measurement was affected by other variables such as mechanical creep in the test structure, thermal effects (e.g self-heating), or potential affects induced by the measuring equipment (such as large time-constants in the parallel ohmmeter), amplifier drift, or other small, unobservable changes in the system.

Several design adjustments can be used to reduce this behavior. First, the rail voltage can be reduced to minimize self-heating. Second, the bridge can be improved from a quarter-bridge to a full-bridge configuration, which provides better thermal compensation for the differential output. Third, low tolerance resistors can be used to improve stability. In addition, warping in the interfacing surfaces was observed as a result of the heating element, suggesting the mechanical variability during measurement.

Although the resulting output produced a usable dataset for characterization, testing with an alternate strain gauge would be necessary to determine whether the observed behavior originated from the gauge itself or the amplifying system.

B. Feedback to maintain a balanced bridge with a quarter bridge

Refer to the following subsections for specific answers related to feedback.

1) Type of signal to feedback to maintain a balanced bridge: The feedback signal used to maintain a balanced bridge would be the differential output voltage of the Wheatstone bridge. This voltage represents the error signal that indicates the imbalance between the bridge sides. When the bridge is balanced the differential voltage is zero (ideal). Any deviation from the zero output signals the resistance of one of the bridge arms has changed as a result of strain. A feedback loop could be implemented to use the error signal to adjust the bridge elements until the output returns to its balanced condition.

2) Systems that implement feedback?: Feedback bridge balancing is used in several measurement systems. In these applications the bridge output is used as an error signal in a closed-loop control system that adjusts a compensating element (such as a transistor or thermistor to control resistance) within the bridge to maintain balance and reduce drift caused by temperature changes, component aging, or long-term instability.

C. Theory

1) Suppose that the bridge was implemented using 1 force sensor, 2 resistors, and the $2nd$ force sensor was replaced with a component ($R_{\Delta T}$) designed to approximate the temperature co-efficient of the strain gauge, but $R_{\Delta T}$ is only a good approximation for T<50C. Then suppose the circuit was placed in a thermally conductive container and affixed to an industrial machine that generates applies 80C to the outside of the container. How might you design a feedback system inside the instrumentation container to help keep the bridge balanced?: In this bridge, the thermistor compensates such that temperature changes do not produce a differential output. Below 50°C, the thermistor and strain gauge track closely enough so both arms of the bridge change at approximately the same rate, allowing the voltage division to remain balanced. However, above 50°C the resistance of the strain gauge increases at a different rate than the thermistor, causing the bridge to produce an error voltage even when no strain is applied.

A potential feedback solution is to use this birdge output as the offset voltage in the amplifier stage. The differential bridge output is monitored and passed through a, slowed, feedback path so only constant thermal mismatch is corrected. The differential signal is then applied as an offset to the amplifier input or reference node, driving the amplifier output back toward zero under zero-strain conditions.

This method keeps the thermistor as the primary compensator, while the feedback loop corrects the additional offset that appears once the strain gauge and thermistor no longer track each other accurately at elevated temperature.

D. Conclusion

The design and characterization of the force-sensing instrumentation system successfully demonstrated the integration of a Wheatstone bridge with active signal conditioning. Experimental results confirmed that the system achieved a stable calibration curve, with the most reliable linear performance observed at loads below 200 g. While the instrumentation amplifier provided the necessary DC offset and high input impedance to preserve signal integrity, the system remained susceptible to external variables.

Significant thermal sensitivity was observed, with an average temperature coefficient of 1.37%/°C, underscoring the limitations of a quarter-bridge configuration in environments with fluctuating temperatures. Analysis of the CMRR showed that the total system rejection is dominated by the bridge's inherent imbalance rather than the amplifier's performance.

Additionally, although the system demonstrated the ability to detect 1 g and 5 g load variations, mechanical settling time and sensor drift were identified as primary factors limiting absolute precision. Future iterations utilizing a full-bridge configuration or active thermal feedback loops, as discussed, would likely enhance stability and minimize the observed drift, providing a more robust solution for high-accuracy load measurement.

Acknowledgment

Dr. Tye Gardner